

ADVANCED ROUTING ARCHITECT: EXPERT CURRICULUM v3.0

MODULE 1: OSPFV2 ADVANCED TUNING & SCALABILITY

OSPF stability in large enterprises relies on LSDB management. Using LSA Throttling and Incremental SPF ensures the CPU is not overwhelmed during flapping. Area types (Stub, NSSA) are mandatory to minimize inter-area flooding.

```
router ospf 1
 timers throttle spf 50 200 5000
 timers throttle lsa all 0 500 5000
 area 10 nssa no-summary
 auto-cost reference-bandwidth 100000
```

MODULE 2: EIGRP NAMED MODE & WIDE METRICS

EIGRP Named Mode is the unified configuration format. It supports Wide Metrics (64-bit) for 10G+ interfaces and native SHA-256 authentication for the Control Plane.

```
router eigrp MY_ENT_NETWORK
 address-family ipv4 autonomous-system 100
  af-interface default
   authentication mode hmac-sha-256 CISCO_KEY
   hello-interval 5
   hold-time 15
 topology base
 variance 2
 redistribute static
```

MODULE 3: BGP POLICY & ADVANCED TRAFFIC ENGINEERING

Managing multi-homed ISP connections requires BGP Path Attributes manipulation. Local Preference controls exit paths, while AS-Path Prepend and Communities influence inbound traffic from the Internet.

```
router bgp 65001
 address-family ipv4
  neighbor 10.1.1.2 route-map SET_LP_IN in
  !
 route-map SET_LP_IN permit 10
  set local-preference 300
  set community 65001:100
```

MODULE 4: VRF-LITE & MULTI-TENANT ISOLATION

Virtual Routing and Forwarding (VRF) creates independent routing tables. This is crucial for security (isolating Guest vs Internal) and supporting overlapping IP ranges in a shared infrastructure.

```
vrf definition SECURE_ZONE
 rd 100:10
 address-family ipv4
 !
 interface Gi0/1.10
  vrf forwarding SECURE_ZONE
 ip address 192.168.10.1 255.255.255.0
```

MODULE 5: ADVANCED ACLS: TIME-BASED & EXTENDED FILTERING

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Expert security implementation uses more than just static rules. Time-based ACLs allow access only during business hours, and Object-Groups make large ACLs easier to manage and audit.

```
time-range WORK_HOURS
  periodic weekdays 08:00 to 17:00
!
ip access-list extended ACL_RESTRICT_ACCESS
  permit tcp 192.168.10.0 0.0.0.255 any eq 443 time-range WORK_HOURS
  deny ip any any log
```

MODULE 6: SAFE REDISTRIBUTION WITH ROUTE-TAGGING

To prevent routing loops in multi-point redistribution, we must use Route Tags. This marks external routes so they aren't redistributed back into their source protocol.

```
route-map TAG_OSPF_TO_EIGRP permit 10
  set tag 110
!
router eigrp MY_ENT_NETWORK
  address-family ipv4 as 100
  topology base
  redistribute ospf 1 route-map TAG_OSPF_TO_EIGRP
```